

Assignment - 1

Q1. $A = \{-1, 0, 1, 2\}$

$B = \{1, 2, 3, 4\}$

a) $A \cup B = \{-1, 0, 1, 2, 3, 4\}$

b) $A \cap B = \{1, 2\}$

c) $A \setminus B = \{-1, 0\}$

d) $\overline{A} \cap B = \{3, 4\}$

Q2. $P_a(H) = 0.6$ $P_a(T) = 0.4$

$P_b(H) = 0.4$ $P_b(T) = 0.6$

a) $\Omega = \{HH, HT, TH, TT\}$

b) $X = \{0, 1, 2\}$

↓

no of head

$P(E_1) = P(X=1) + P(X=2)$

↓

$H_a T_b$

+

$H_b T_a$

↓

$(0.6)(0.4)$

0.24

$(0.6)(0.6) + (0.4)(0.4)$

$0.36 + 0.16$

0.52

$P(E_1) = 0.52 + 0.24$

$= 0.76$

$$E_2 \rightarrow H_a T_b$$

$$P(E_2) = P(X=1 \text{ out } H_2) = \underline{0.36}$$

$$C \rightarrow F(\text{event space}) = \left\{ \begin{array}{l} H_a H_b \\ H_a T_b \\ T_a H_b \\ T_a T_b \end{array} \right\}$$

$$P(H_a H_b) = 0.24$$

$$P(H_a T_b) = 0.36$$

$$P(T_a H_b) = 0.16$$

$$P(T_a T_b) =$$

$$P(T_a T_b) = 0.24$$

$$P(f) = 0.24 + 0.24 + 0.36 + 0.16 = 1$$

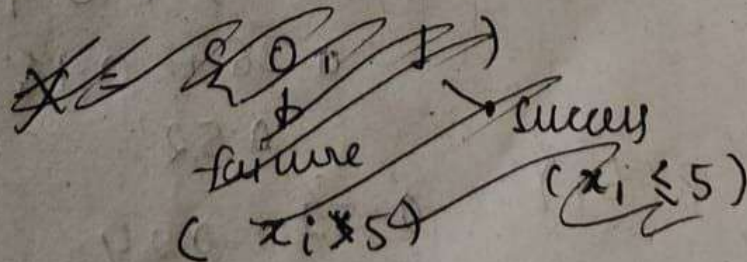
Q 2.2

$$\Omega = \{x_1, x_2, x_3, x_4, x_5\} \quad x_i \in R$$

$$1 \leq i \leq 5$$

$$\omega = \{5.76, 4.56, 2.33, 6.32, 5.4\}$$

ω is one of them, one of element



$$\Omega_X = \{0, 1, 2, 3, 4, 5\}$$

↓
No. of success

describe X as $\omega = \{x_1, x_2, x_3, x_4, x_5\}$

$$I(x_i) = \begin{cases} 1 & x_i \leq 5 \text{ success} \\ 0 & x_i > 5 \text{ failure} \end{cases}$$

always
define

X as function

$$X(\omega) = \sum_{i=1}^5 I(x_i)$$

$$X \text{ for given data} = 0 + 1 + 1 + 0 + 0 = 2$$

Sample space for X :

$$\Omega_X = \{0, 1, 2, 3, 4, 5\}$$

⑤

$$P(\text{success}) = p$$

$X \sim$ as binomial random variable

$$P(X=k) = {}^5C_k p^k (1-p)^{5-k}$$

$$k = 0, 1, 2, 3, 4, 5$$

probability distribution Table.

k	$P(X=k)$
0	${}^5C_0 (1-p)^5$
1	${}^5C_1 p (1-p)^4$
2	${}^5C_2 p^2 (1-p)^3$
3	${}^5C_3 p^3 (1-p)^2$
4	${}^5C_4 p^4 (1-p)$
5	${}^5C_5 p^5$

Q. 3

1) $f(k, n, p)$

$$= \frac{n!}{k! (n-k)!} p^k (1-p)^{n-k}$$

$$k = 45$$

$$n = 50$$

$$p = 0.9$$

$$p(X=45) = \frac{50}{45! 5!} p^{45} (1-p)^5$$

$$p(X=45) \approx 0.184$$

2) $f(k, \lambda) = \frac{\lambda^k e^{-\lambda}}{k!}$ $\lambda = 10$

$$P(X=0) = \frac{10^0 e^{-10}}{0!} = e^{-10}$$

$$p(X=0) \approx 0.0000454$$

Q 4.

Q.

$$P(X=k) = f(k) = \frac{10^k e^{-10}}{k!}$$

No. of
infection
in 1 day

$$4.1 = P(X=100) = \frac{10^{100} e^{-10}}{100!} \approx 0$$

$$4.2 = P(5 \leq X \leq 15) = \sum_{k=5}^{k=15} \frac{10^k e^{-10}}{k!} \approx 0.829$$

$$4.3 \rightarrow \Omega = \{0, 1, 2, 3, \dots\}$$

4.4 →

4.5 → 10 days infection person

18, 20, 23, 25, 22, 30, 11, 10, 23, 12

$$\text{Mean of observed data} = \bar{x} = \frac{194}{10} = 19.4$$

but $\lambda = 10$

∴ On average infected person

∴ ~~the~~ ~~researcher's~~ claim is very likely incorrect.

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[1] 11 12  8  3 12
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[1]  8 10  7 14 10
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Poisson PMF with lambda = 10

